

Dana's work when solving the equation $x^2 + 4x = 3$ by completing the square is shown. Use Dana's work to answer questions 1 and 2.

$$x^2 + 4x = 3$$

$$(x^2 + 4x) - 3 = 0$$

$$(x^2 + 4x + 4) - 3 - 4 = 0$$

$$(x + 2)^2 - 1 = 0$$

$$(x + 2)^2 = 1$$

$$x + 2 = \pm 1$$

$$x + 2 = 1 \text{ or } x + 2 = -1$$

$$x = -1 \text{ or } x = -3$$

$$(x + 2)^2 - 7 = 0$$

$$(x + 2)^2 = 7$$

$$(x + 2)^2 = \pm\sqrt{7}$$

$$x = -2 \pm \sqrt{7}$$

1. Circle the step where Dana made an error.
2. Correct Dana's error and the steps that follow to find the correct solution(s) to the equation. Use the space provided above to show the correct steps.

Caleb's work when solving the equation $3x^2 + 36x = -18$ by completing the square is shown. Use Caleb's work to answer questions 3 and 4.

$$3x^2 + 36x = -18$$

$$3x^2 + 36x + 18 = 0$$

$$3(x^2 + 12x) + 18 = 0$$

$$3(x^2 + 12x + 36) + 18 - 36 = 0$$

$$3(x + 6)^2 - 18 = 0$$

$$3(x + 6)^2 = 18$$

$$(x + 6)^2 = 6$$

$$x + 6 = \sqrt{6} \text{ or } x + 6 = -\sqrt{6}$$

$$x = -6 + \sqrt{6} \text{ or } x = -6 - \sqrt{6}$$

$$\frac{3(x+6)^2}{3} - \frac{90}{3} = 0$$

$$(x + 6)^2 - 30 = 0$$

$$(x + 6)^2 = 30$$

$$x + 6 = \pm\sqrt{30}$$

$$x + 6 = \pm\sqrt{30}$$

$$x = -6 \pm \sqrt{30}$$

3. Circle the step where Caleb made an error.
4. Correct Caleb's error and the steps that follow to find the correct solution(s) to the equation. Use the space provided above to show the correct steps.

Solve each equation by completing the square. Show your work.

5. $x^2 + 2x - 9 = 21$

$$(x^2 + 2x + 1) - 9 - 21 - 1 = 0$$

$$(x + 1)^2 - 31 = 0$$

$$(x + 1)^2 = 31$$

$$x + 1 = \pm\sqrt{31}$$

$$x = -1 \pm \sqrt{31}$$

6. $x^2 - 6x + 4 = 2$

$$(x^2 - 6x + 9) + 4 - 2 - 9 = 0$$

$$(x - 3)^2 - 7 = 0$$

$$(x - 3)^2 = 7$$

$$x - 3 = \pm\sqrt{7}$$

$$x = 3 \pm \sqrt{7}$$

7. $0 = 3x^2 - 18x + 6$

$$3(x^2 - 6x + 9) + 6 - 27 = 0$$

$$3(x - 3)^2 - 21 = 0$$

$$(x - 3)^2 - 7 = 0$$

$$(x - 3)^2 = 7$$

$$x - 3 = \pm\sqrt{7}$$

$$x = 3 \pm \sqrt{7}$$

8. $2x^2 + 20x = -28$

$$2(x^2 + 10x + 25) + 28 - 50 = 0$$

$$2(x + 5)^2 - 22 = 0$$

$$(x + 5)^2 - 11 = 0$$

$$(x + 5)^2 = 11$$

$$x + 5 = \pm\sqrt{11}$$

$$x = -5 \pm \sqrt{11}$$

9. $x^2 - 18x + 80 = 0$

$$(x^2 - 18x + 81) + 80 - 81 = 0$$

$$(x - 9)^2 - 1 = 0$$

$$(x - 9)^2 = 1$$

$$x - 9 = 1 \quad x - 9 = -1$$

$$x = 10 \quad x = 8$$

10. $-23 = x^2 - 12x$

$$(x^2 - 12x + 36) + 23 - 36 = 0$$

$$(x - 6)^2 - 13 = 0$$

$$(x - 6)^2 = 13$$

$$x - 6 = \pm\sqrt{13}$$

$$x = 6 \pm \sqrt{13}$$